Satellite Ground Station Time Sharing

This document defines the principal stakeholder goals and business functions of the Satellite Ground Station Time Sharing system. It focuses on observable requirements and avoids prescribing implementation details.

1. Stakeholder Context

Stakeholder category	Stakeholder	Role	Primary interest
Internal - Operational	Eng. Dina Hosny	Station Operations Lead	Reliable passes, station operation, pass plans, and scheduling changes.
Internal - Operational	Eng. Ahmed Kamel	Frequency Coordinator	Valid frequency licensing before scheduling.
Internal - Executive	Dr. Yasser Elnaggar	Ground Segment Manager / Sponsor	Fair use of limited station time and resource allocation.
Internal - Executive	Dr. Samir Ezzat	Head of Information Security	Secure remote commanding and prevention of unauthorized control.
External - Operational	Dr. Lamia Rashad	Mission Operator, Cairo University CubeSat	Contact requests, passes, telemetry, and remote commanding.
External - Operational	Mr. Sani Bello	Mission Operator, Partner University in Nigeria	Ground-station access and clear local-time scheduling.
External - Executive	University / Mission Management	Participating universities and missions	Fair access, resource visibility, and scheduling outcomes.

2. User Stories

The following stories express stakeholder goals using the standard user-story form. Each story is intentionally limited to one principal goal so that it can be estimated, implemented, and tested independently.

ID	Stakeholder category	Stakeholder / role	User story
US01	External - Operational	Dr. Lamia Rashad - Mission Operator	As a mission operator, I want to submit a contact request with the required passes and duration, so that my satellite can be considered for ground-station scheduling.
US02	External - Operational	Mr. Sani Bello - Mission Operator	As a mission operator, I want to view scheduled pass times in my local time zone, so that I can plan my mission activities without manually converting UTC.
US03	Internal - Operational	Eng. Dina Hosny - Station Operations Lead	As a station operations lead, I want to receive the confirmed contact information needed for a pass, so that the ground station can prepare and execute the contact reliably.
US04	Internal - Operational	Eng. Ahmed Kamel - Frequency Coordinator	As a frequency coordinator, I want the system to verify a satellite's frequency license for the pass date, so that satellites without a valid license are not scheduled.
US05	Internal - Executive	Dr. Yasser Elnaggar - Ground Segment Manager / Sponsor	As the ground segment manager, I want competing requests for limited station time to be allocated according to the established priority rules, so that shared ground-station resources are used fairly.
US06	Internal - Executive	Dr. Samir Ezzat - Head of Information Security	As the head of information security, I want remote commands to be checked against the intended satellite before transmission, so that a command is not sent to the wrong satellite.
US07	External - Executive	University / Mission Management	As university or mission management, I want to view my university's ground-station resource usage, so that I can monitor how the shared resource is being used.
US08	External - Executive	University / Mission Management	As university or mission management, I want to know the outcome of a scheduling request, so that I can understand whether the requested station time was granted or rejected.

3. Acceptance Criteria

Acceptance criteria explain what should happen for each user story. They are written using the Given-When-Then format

Story	Given	When	Then
US01	A mission requires ground-station contact time.	The mission operator submits a contact request.	The request is recorded with the required pass and duration information.
US01	A contact request has been submitted.	The request enters scheduling.	The request is available for consideration according to the scheduling rules.
US02	A pass has been scheduled.	The mission operator views the schedule.	The pass time is displayed in the operator's local time zone.
US02	The operator needs to plan a mission activity.	The operator reviews an upcoming pass.	The displayed time can be used without manual UTC conversion.
US03	A contact has been confirmed.	The station operations lead accesses the contact information.	The information needed to prepare and execute the contact is available.
US03	A scheduled contact is approaching.	The station operations lead prepares for the contact.	The confirmed contact information can be used as the operational basis for the pass.
US04	A satellite is being considered for a pass.	The system checks the satellite's license.	The license status is evaluated for the pass date.
US04	The license is invalid for the pass date.	Scheduling eligibility is determined.	The satellite is not scheduled for that pass.
US05	Two or more missions compete for limited station time.	The system resolves the scheduling conflict.	The established priority and conflict-resolution rules determine which request receives the contested slot.
US05	A satellite has lost a contact because of a conflict.	A later conflict is resolved.	The defined anti-starvation rule is applied so the same satellite is not repeatedly disadvantaged.
US06	An authorized operator is preparing a remote command.	The command is checked before transmission.	The target satellite is verified before the command is sent.
US06	The command targets a satellite that is not the satellite in view.	The system performs the target check.	The command is not transmitted to the wrong satellite.
US07	A university has used shared ground-station resources.	Management views resource usage.	The system displays the university's relevant resource/credit usage.
US07	Management reviews resource consumption.	The usage information is displayed.	The information allows management to monitor the university's share of the shared resource.
US08	A scheduling request has been processed.	Management views the request outcome.	The outcome is visible as granted or rejected.
US08	A request is rejected because another satellite receives the contested slot.	The rejection is displayed.	The system identifies the satellite that received the contested slot.

4. INVEST Evaluation

INVEST is used here as a quality check for the user stories: It stands for Independent, Negotiable, Valuable, Estimable, Small, and Testable. A triangle means the story is acceptable but should be monitored for future decomposition.

ID	I	N	V	E	S	T	Reason
US01	✓	✓	✓	✓	✓	✓	Clear, focused request-submission goal.
US02	✓	✓	✓	✓	✓	✓	Single goal: viewing scheduled passes in local time.
US03	✓	✓	✓	✓	✓	✓	Focused on making confirmed contact information available for operations.
US04	✓	✓	✓	✓	✓	✓	Single eligibility check with an observable result.
US05	✓	✓	✓	✓	△	✓	Still acceptable, but the detailed fairness rules could be split later if the rules become too large.
US06	✓	✓	✓	✓	✓	✓	Single safety goal: verify the intended satellite before transmission.
US07	✓	✓	✓	✓	✓	✓	Single management-visibility goal.
US08	✓	✓	✓	✓	✓	✓	Single goal: understand the scheduling outcome.

- **✓:** The user story meets the INVEST criterion.
- **△:** The user story partially meets the criterion, but it may need improvement or further division (to smaller user stories).

5. Event Decomposition

Events are grouped into external, temporal, or state events. Each event is linked to the business function that responds to it.

Event ID	Event	Type	Trigger	Business response
E01	Mission operator submits a contact request.	External	Mission operator	Submit Contact Request
E02	The scheduled time for preparing a pass is reached.	Temporal	Scheduled time	Prepare Pass / Contact
E03	A proposed pass has a license-validity condition.	State	License status	Check Frequency License
E04	The scheduled daily orbit-data update time is reached.	Temporal	Scheduled time	Calculate Satellite Passes
E05	Two requests compete for the same unavailable station time.	State	Scheduling conflict	Resolve Scheduling Conflict
E06	A scheduled contact is completed and telemetry is received.	External	Contact completion	Store Telemetry
E07	An authorized operator initiates a remote command during an applicable pass.	External	Authorized mission operator	Execute Remote Command
E08	The weather condition affecting a scheduled pass changes.	State	Weather condition	Handle Weather Risk

6. Business Use Cases

These use cases represent the principal business functions derived from the stakeholder goals and system events.

ID	Use case	Primary actor / user	Business goal
UC01	Submit Contact Request	Mission Operator	Request ground-station communication time.
UC02	View Scheduled Passes	Mission Operator	View upcoming scheduled pass times in the appropriate local time.
UC03	Prepare Pass Plan	Station Operations Lead	Prepare the information required to execute a scheduled contact.
UC04	Check Frequency License	Frequency Coordinator	Ensure that the satellite is eligible for the proposed pass.
UC05	Calculate Satellite Passes	Scheduling Function	Calculate upcoming pass opportunities from current orbit information.
UC06	Resolve Scheduling Conflict	Station Operations Lead / Scheduling Function	Resolve competing requests using the defined scheduling rules.
UC07	Allocate Ground-Station Time	Ground Segment Manager / Sponsor	Manage allocation of limited station time among missions.
UC08	Store Telemetry	System / Mission Operator	Store received telemetry for access by the satellite owner.
UC09	Execute Remote Command	Authorized Mission Operator	Send an authorized command to the intended satellite during an applicable pass.
UC10	Handle Weather Risk	Station Operations Lead	Identify weather-related operational risk and support the operational response.
UC11	Monitor Resource Usage	University / Mission Management	View the university's shared ground-station resource usage.
UC12	Explain Scheduling Outcome	University / Mission Management	Provide the result of a scheduling request and relevant rejection information.

7. Perfect Technology Assumption

This analysis focuses on the actual business functions and stakeholder goals, rather than features that only deal with technical limits or routine administration. For this reason, authentication, logout, password changes, database backup, and database restoration are not listed as separate business use cases at this stage. Security is included when it is needed to support safe remote commanding.

8. Final Quality Check

Check	Result
User stories	All stories use a consistent As a / I want / so that structure and express stakeholder value.
Story size	Stories are kept to one principal goal. US05 is the only one flagged for possible future decomposition if detailed fairness rules expand.
Acceptance criteria	Each story has observable criteria rather than vague statements.
INVEST	The evaluation is presented as a quality assessment of the stories, not as a requirement itself.
Events	External, temporal, and state events are separated and connected to business responses.
Use cases	Use-case names describe business actions rather than technical components.

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Brief Use Case Descriptions

Description convention: Each brief description summarizes the main steps of each use case.

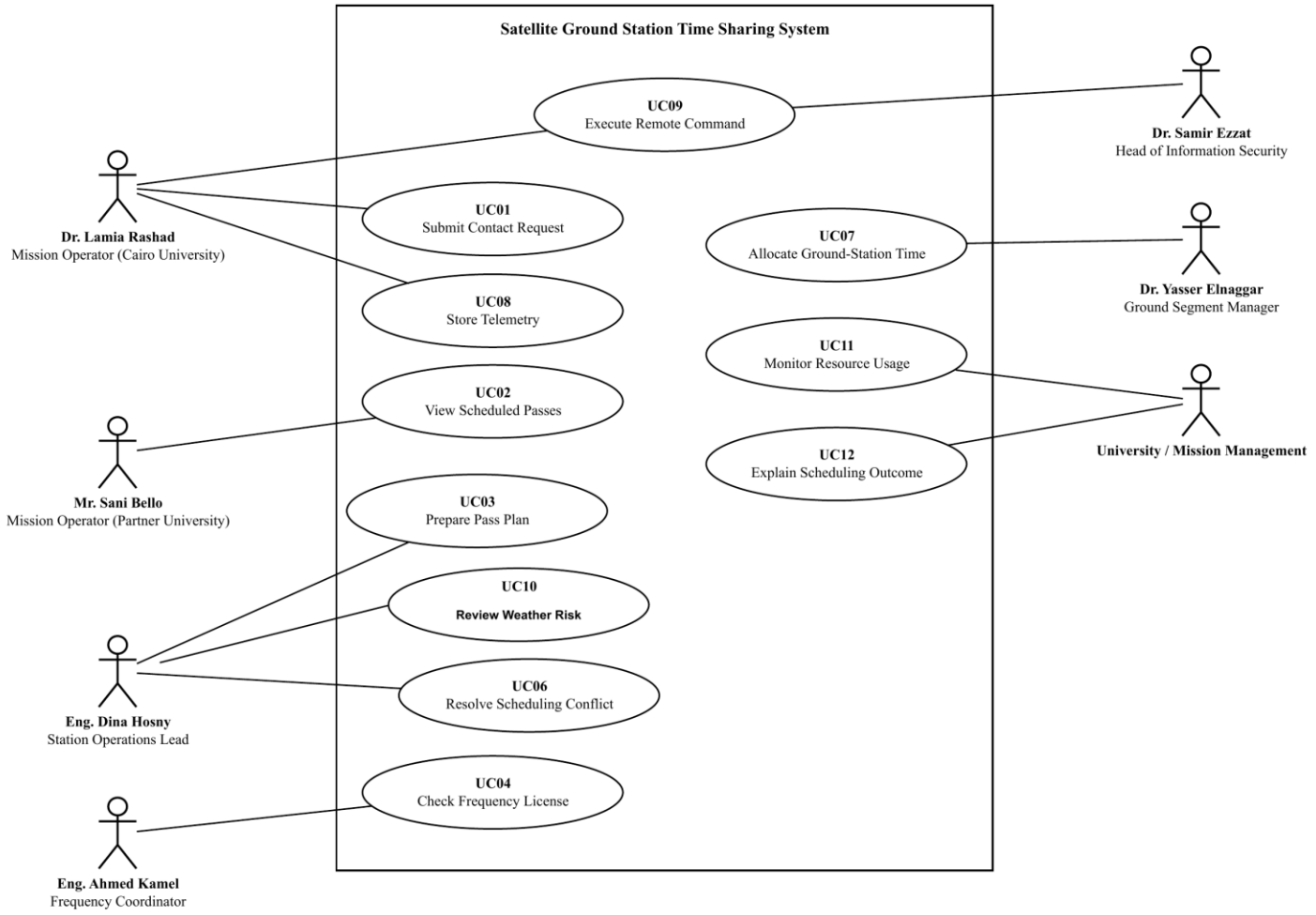
Use Case	Brief Use Case Description
UC01 - Submit Contact Request	A Mission Operator starts this use case when a satellite requires ground-station contact time; the operator submits a request identifying the satellite and specifying the required minimum passes per day, minimum contact duration, and any mandatory passes, and the system records the request for scheduling.
UC02 - View Scheduled Passes	A Mission Operator starts this use case when they need to review upcoming contacts; the operator requests the satellite's schedule, and the system displays the confirmed pass dates and times in the appropriate local time so the operator can plan for each contact.
UC03 - Prepare Pass Plan	The Station Operations Lead starts this use case when a scheduled contact is approaching; the lead prepares the pass plan, including the contact details, antenna-pointing angles, and commands, and the system makes the plan available to the remote station at least 30 minutes before the contact so it can execute the received plan even if the link to the center later fails.
UC04 - Check Frequency License	The Frequency Coordinator starts this use case when a satellite is being considered for a proposed pass; the system checks whether the satellite's frequency license is valid on the date of that pass, and the coordinator uses the result to determine whether the satellite is eligible to be scheduled.
UC05 - Calculate Satellite Passes	At the scheduled daily orbit-data update, the system obtains fresh TLE data and calculates satellite visibility opportunities over the ground stations for the next seven days; the Scheduling Function uses these calculated passes as input to contact scheduling, with the stated performance target of completing calculations for 15 satellites across 7 stations in under 10 minutes.
UC06 - Resolve Scheduling Conflict	The Station Operations Lead and Scheduling Function start this use case when two requests compete for the same unavailable station time; the system supports resolving the conflict first by moving a contact to another station, next by splitting a pass if both operators accept the shorter contact, and finally by applying mission priority, while giving the satellite that loses a contact higher priority next time to reduce repeated loss of access.
UC07 - Allocate Ground-Station Time	The Ground Segment Manager / Sponsor starts this use case when ground-station capacity must be allocated among participating universities and missions; the manager oversees allocation using the annual allowance of 600 credits per university and the rule that one minute of contact costs one credit, while accounting for the 50 free passes granted to a new science mission during its first 30 days and replacement contacts owed when a pass is missed because of the station.
UC08 - Store Telemetry	This use case is triggered when a scheduled contact is completed and telemetry is received; the system stores the telemetry files and associates them with the relevant satellite and contact so that the satellite owner can access and download them, with telemetry retained for one year.
UC09 - Execute Remote Command	An Authorized Mission Operator starts this use case during the university's own scheduled pass by submitting a remote command; the system checks the required authorization conditions, verifies that the target satellite ID matches the satellite in view, requests confirmation before sending, and logs the command, while allowing operators to block remote commanding immediately.

Use Case	Brief Use Case Description
UC10 - Review Weather Risk	The Station Operations Lead starts this use case when a weather condition affecting a scheduled pass changes or a dust-storm risk is flagged; the system identifies the affected pass and presents the risk information, and the operator decides whether to keep the pass.
UC11 - Monitor Resource Usage	University / Mission Management starts this use case when it needs to review its use of shared ground-station resources; the representative requests the usage information, and the system presents the university's pass usage and credit consumption so management can monitor its allocation.
UC12 - Explain Scheduling Outcome	This use case is triggered when a contact request receives a scheduling decision; the system communicates the outcome to the requesting mission operator or university representative, and, when a request is rejected because of a conflict, identifies the satellite that received the contested slot so the decision is understandable.

Use Case Diagram

Satellite Ground Station Time Sharing System

Use Case Diagram – All Use Cases



Satellite Ground Station Time Sharing System

Use Case Diagram – External and Internal Use Cases

